

RAVINDRA SINGH

Data Scientist • Machine Learning Engineer • Embedded / Edge AI

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PROFESSIONAL SUMMARY

Data Scientist and Machine Learning Engineer with 3+ years of experience designing, deploying, and optimizing AI models for speech recognition, wake-word detection, computer vision, sensor-based activity recognition, and embedded/edge AI. Skilled in building lightweight neural networks for resource-constrained devices using TensorFlow Lite, model quantization, and low-latency inference. Shipped 10+ production models at 92–98% accuracy on embedded hardware.

TECHNICAL SKILLS

Core: Python, Machine Learning, Deep Learning, NLP, Generative AI, LLMs, SLMs

Deep Learning: CNN, RNN, LSTM, GRU, ANN, Conv1D/Conv2D, Transformers, MFCC

Frameworks & Libraries: TensorFlow, Keras, PyTorch, Scikit-Learn, FastAPI, Pandas, NumPy, Matplotlib, Seaborn, OpenCV, Git

Embedded / Edge AI: GPX10, TensorFlow Lite, Model Quantization & Optimization, Real-Time Inference, IMU Sensors, ESP32 Camera

WORK EXPERIENCE

Data Scientist | Ambient Scientific

May 2023 – Present | Bangalore, India

- Developed and deployed 10+ AI models (wake-word detection, speech recognition, computer vision, HAR, fall detection) on GPX10 embedded platforms at 92–98% accuracy.
- Designed low-resource neural network architectures (5K–43K parameters) and cut memory footprint and latency using TensorFlow Lite and model quantization.
- Led data collection, preprocessing, augmentation, feature extraction (MFCC), and EDA in Python across speech and sensor datasets.
- Evaluated models with accuracy, precision, recall, F1-score, and confusion-matrix analysis; monitored and tuned performance in production.
- Partnered with cross-functional product teams to align AI solutions with hardware and product requirements.

Machine Learning Engineer | Neelavath Software Solution

2022 – 2023 | Coimbatore, India

- Built predictive ML models and implemented deep learning architectures (CNN, RNN, LSTM, GRU, ANN) to improve business insights and decision-making.
- Performed data cleaning, feature engineering, and dataset preparation for machine learning pipelines.
- Deployed and maintained models in production, improving reliability through systematic validation and testing.

KEY PROJECTS

- Voice Signature (Speaker Verification)** — Built end to end, from data collection and cleaning (23K+ speakers) to deployment; ~250K-parameter TDNN speaker-embedding model with INT8 quantization-aware training; 1.04% EER.
- Wake-Word Detection (Alexa, Neura, Amigo)** — GRU/CNN keyword-spotting models (~5K parameters) using MFCC features; 98% accuracy with low-latency real-time inference on GPX10.
- Multi-Command Recognition** — Embedded speech-command models (Conv1D/Conv2D + MFCC) for accurate multi-command classification on resource-constrained devices.
- Fall Detection** — 43K-parameter IMU model (accelerometer $\pm 16G$, gyroscope 2000 dps; 1.7s window, 26 Hz ODR); 96% accuracy on GPX10.
- Human Activity Recognition (HAR)** — Motion-classification models from accelerometer and gyroscope IMU features; 92% accuracy on embedded hardware.
- Face ID Recognition** — Lightweight CNN (10K parameters) on ESP32 Camera imagery; 98% face-recognition accuracy on GPX10.
- Healthcare SLM Chatbot** — Conversational AI using a 1M-parameter Transformer Decoder with custom tokenizer; 82% response accuracy with an edge-deployable footprint.

EDUCATION

Master of Computer Applications (MCA) | SNS College of Technology

2021 – 2023 | Coimbatore, India

Bachelor of Science (B.Sc), Maths / Chemistry / Physics | Brij University

2015 – 2018 | Bharatpur, India

CERTIFICATIONS & LANGUAGES

Generative AI & LLMs — Coursera (2025) • Machine Learning — Internshala (2022) • Data Science Foundation — Great Learning (2021)

Languages: English — Full Professional (C1) • French — Basic (A1)